Sovereignty Unchained

Next-Gen Decentralized Sandbox Game Architecture

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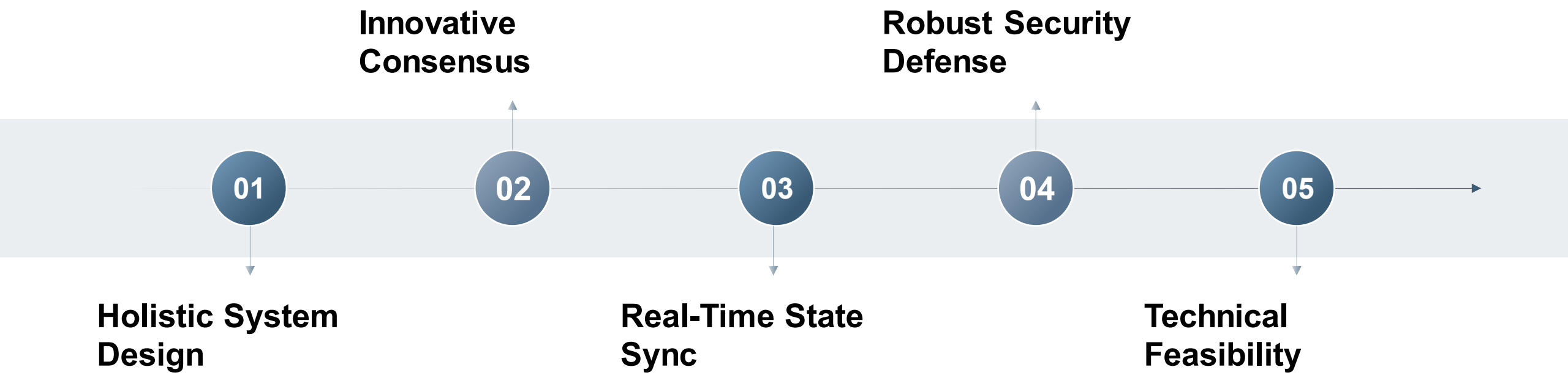
Agenda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Holistic System Design

This chapter introduces the core architectural principles and layered design of "Sovereignty Unchained," laying the foundation for a serverless, player-owned gaming ecosystem.

Layered Architecture

The system is composed of four distinct layers, each with a specific role in maintaining the game world.

1 Application Layer

Provides rich game functionalities including character customization, sandbox building, social systems, economic transactions, and family legacy features.

2 Synchronization Layer

Responsible for real-time state synchronization and conflict resolution across all nodes using CRDT data types and the Gossip protocol.

3 Network Layer

Built on libp2p, this layer forms a P2P network and efficiently manages node discovery and resource allocation via the GossipSub protocol.

4 Trust Layer

Establishes a decentralized trust foundation by integrating distributed trust roots with TPM hardware anchoring and DID identity systems.

Core Architectural Principles

Four fundamental principles define the next-generation nature of "Sovereignty Unchained."

1

Decentralization

All player devices act as both clients and servers, directly communicating and collectively maintaining the game world's global state without a central server.

2

Perpetual Existence

The serverless architecture eliminates the risk of shutdown due to business decisions or technical failures, ensuring the game world exists in perpetuity.

3

Player Sovereignty

Player data is encrypted and stored locally, with identities recoverable through social mechanisms, ensuring true ownership of assets and data privacy.

4

Immutable Rules

Core game rules are codified in tamper-proof code and protected by nine distributed trust roots, preventing arbitrary rule changes.

Innovative Consensus

This chapter analyzes the limitations of traditional blockchain consensus and details the "Social Witness" mechanism, a novel consensus design tailored for gaming.

Limitations of Traditional Consensus

Traditional mechanisms like PoW, PoS, and PBFT are unsuitable for real-time gaming scenarios.

1

Proof of Work (PoW)

High energy consumption and large transaction confirmation delays make it impossible to meet the low-latency requirements of real-time game state synchronization.

2

Proof of Stake (PoS)

Vulnerable to manipulation by "whales," where voting power is determined by token holdings, violating the principle of game fairness.

3

Practical BFT (PBFT)

Limited by the number of nodes, it suffers from severe scalability issues, making it inadequate for a large-scale player network.

Social Witness Mechanism

A consensus mechanism that leverages the social relationship network within the player community.

1 Core Principle

Any rule change request must first be verified by a distributed trust root, then approved by a 5/9 majority vote before a state update is triggered.

2 Distributed Trust Root Structure

A "triple-core + six-peripheral" mirror architecture is used. Core mirrors are stored in secure device firmware areas for emergency recovery, while peripheral mirrors are distributed across the P2P network for daily validation.

3 Voting Mechanism

A rule change requires the approval of a 5/9 majority of nodes. Nodes independently verify requests, and voting results are propagated via the Gossip protocol, ensuring high security.

DID Identity Binding

Binding player identities (DIDs) to their social networks to calculate witness weights.

01

DID Generation

After completing a strong real-name verification (e.g., facial recognition), the system generates a unique, decentralized identity identifier for the player.

02

Key Management

The player's private key is self-custodied, while the public key and DID are recorded on a distributed ledger, ensuring identity security.

03

Social Recovery

Players pre-define multiple "guardians." If an account is lost, it can be recovered with the assistance of a majority of these guardians.

04

Witness Weight

Based on historical behavior records within the player's social network, witness weights are dynamically adjusted to prevent Sybil attacks.

Real-Time State Sync

This chapter outlines the challenges of state synchronization in a P2P network and explains how CRDT and Gossip protocols enable millisecond-level real-time sync.

Synchronization Challenges

1 Network Partitioning

Player devices may go offline or experience network interruptions at any time, requiring state consistency to be guaranteed after partition recovery.

3 Concurrent Conflicts

Multiple players may simultaneously modify the same game state (e.g.,争夺 resources), requiring an effective conflict resolution mechanism.

2 High Latency

Communication delays between global nodes are unpredictable; the synchronization mechanism must be highly tolerant of this latency.

4 Eventual Consistency

The system must ensure that the game state on all nodes eventually converges to the same correct result.

Data synchronization in a P2P game network must solve four core problems.

CRDT Core Application

Conflict-free Replicated Data Types (CRDTs) are used as the core synchronization mechanism.

01

Applicable Types

Different CRDT types (e.g., Last-Writer-Wins Register, G-Set, OR-Set, PN-Counter) are selected based on the specific game state being managed.

02

Synchronization Flow

After a local state change, a corresponding CRDT operation is generated and broadcast via the Gossip protocol. Remote nodes receive and automatically merge these operations, achieving state convergence.

03

Key Properties

CRDTs possess critical properties like commutativity, associativity, and idempotency, ensuring the final state remains consistent regardless of the order or grouping of operations.

Gossip Protocol Optimization

The Gossip protocol is architected and optimized to ensure efficient synchronization in a large-scale network.

1

Protocol Architecture

The GossipSub publish/subscribe model is adopted. Nodes join topic channels of interest, receiving and forwarding only relevant state updates.

2

Performance Optimizations

Techniques include libp2p connection pool management to control per-node connections, interest-based subscription filtering to reduce无效 messages, and adaptive rate limiting to control bandwidth.

3

Fallback Solution

WebRTC pixel streaming is introduced to provide a basic gaming experience for players with low-end hardware.

Robust Security Defense

This chapter analyzes the primary security threats to the system and demonstrates how hardware anchoring and a multi-layered defense system create an impregnable security barrier.

Potential Security Threats

The game network may face several primary attack vectors.

1 Sybil Attack

An attacker creates a large number of fake identities to attempt to manipulate consensus voting or disrupt the economic system.

2 Sybil Cluster

A single entity controls multiple nodes to gain a disproportionate level of influence within the network.

3 Identity Forgery

Impersonating another player's DID identity to steal assets or damage social trust.

4 Rule Tampering

Maliciously modifying the core game code or rules to破坏 game fairness and steal player assets.

Hardware Anchoring Mechanism

Integrating a TPM chip to provide a hardware-level root of trust for each player's device.

01

02

03

TPM Chip Integration

The TPM chip works in concert with the device's firmware partition and hidden partition to form a secure area for triple-core mirror storage.

Anchoring Process

During device registration, the TPM generates a unique device key, which is cryptographically bound to the player's DID. Critical state changes require a signature from the TPM private key to be valid.

Security Properties

This mechanism achieves device uniqueness, key protection, integrity verification, and tamper resistance, ensuring the device's identity is genuine and its state is可信.

Multi-Layered Defense System

Defense in depth is the foundation of a multi-layered security architecture.

1

Identity Layer Defense

Strong real-name verification, DID binding, hardware anchoring, and social recovery mechanisms ensure the authenticity and uniqueness of player identities.

2

Network Layer Defense

A node reputation system is established, per-node connection limits are enforced, and anomalous connection patterns are automatically detected and isolated.

3

Consensus Layer Defense

The 5/9 majority voting principle is upheld, witness weights are dynamically adjusted, and compliance rules are locked in an independent mirror, updated only by community vote.

Continuous Security Audit

A continuous security audit and verification mechanism ensures the system remains secure over time.

01

Code Audit

02

Consensus Verification

03

Compliance Check

04

Penetration Testing

Technical Feasibility

This chapter assesses the project's technical feasibility, presents existing case studies, evaluates technology maturity, and outlines comprehensive risk mitigation strategies.

Existing Case Studies

Small-scale technical validation points have been proven through existing projects.

01

Shaga

Verified the feasibility of a P2P game network, testing direct node-to-node communication and collaboration in a small-scale environment.

02

YOM

Explored decentralized rendering technology, providing an initial technical path for the distributed presentation of the game world.

03

DeepLink

Practiced storing and interacting with game states on-chain, demonstrating the feasibility of blockchain-based game data.

Technology Maturity Assessment

The maturity of the core technology components is evaluated for risk.

1

High Maturity

Technologies like the libp2p network library, CRDT data types, and the GossipSub protocol are well-established and have been widely validated, posing a low risk.

2

Medium Maturity

Technologies like TPM remote attestation and DID social recovery are relatively new but have a solid foundation and are under active development, posing a medium risk.

Risk Mitigation Strategies

Comprehensive strategies are in place to address various potential risks.

01

Technical Risk

•Leverage mature open-source technologies, enforce strict code audits, and design off-chain scaling solutions to handle potential technical bottlenecks.

02

Market Risk

•Adopt a community-consensus-driven economic model to avoid excessive financialization and ensure the game's long-term, healthy development.

03

Governance Risk

•Design voting mechanisms to resist Sybil and whale attacks, and guide the community with professional operations to ensure fair and efficient governance.

04

Compliance Risk

•Establish a dedicated team to actively track global regulatory trends and seek legal advice to ensure the project's compliance worldwide.

Thank You

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